# Tutorial 7

## Quantitative Spatial Economics

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# Indirect utility

We have once derived the indirect utility of worker $\omega$ living in location $i$ and working in location $j$ as:

$$U_{ij\omega} = \frac{z_{ij\omega} B_i w_j Q_i^{\beta-1}}{d_{ij}}, \qquad \text{where } d_{ij} := e^{\kappa\tau_{ij}} \geq 1,$$

where $\tau_{ij}$ is travel time between $i$ and $j$ in minutes, and $\kappa > 0$ scales disutility per minute.

# Location choice

The location choice probability of random worker choosing to live in $i$ and work in $j$ is given by:

$$\pi_{ij} := \frac{\Phi_{ij}}{\Phi} = \frac{T_i E_j \left(d_{ij} Q_i^{1-\beta}\right)^{-\varepsilon} (B_i w_j)^{\varepsilon}}{\sum_{r=1}^{S} \sum_{s=1}^{S} T_r E_s \left(d_{rs} Q_r^{1-\beta}\right)^{-\varepsilon} (B_r w_s)^{\varepsilon}}$$

and after taking the logarithm we get:

$$\ln \pi_{ij} = \underbrace{\ln T_i - \varepsilon(1-\beta) \ln Q_i + \varepsilon \ln B_i}_{\text{residence shifters}} + \underbrace{\ln E_j + \varepsilon \ln w_j}_{\text{workplace shifter}} - \underbrace{\varepsilon \ln d_{ij}}_{=\varepsilon\kappa\tau_{ij}} - \underbrace{\ln \Phi}_{\in\mathbb{R}}. \qquad (*)$$

# Gravity equation

Define:

$$\zeta_i := \ln T_i - \varepsilon(1-\beta)\ln Q_i + \varepsilon \ln B_i, \qquad \eta_j := \ln E_j + \varepsilon \ln w_j, \qquad \nu := \varepsilon\kappa$$

and in that case the equation $(*)$ boils down to:

$$\ln \pi_{ij} = \nu\tau_{ij} + \zeta_i + \eta_j.$$

In that case:

- $\nu$ is the location choice probability semi-elasticity of travel time
- $\zeta_i$ is the origin fixed effect of location $i$
- $\eta_j$ is the destination fixed effect of location $j$

# Reduced form estimation

Based on:

$$\ln \pi_{ij} = \nu \tau_{ij} + \zeta_i + \eta_j,$$

we would like to estimate (at block level):

$$\ln R_{ij} = \nu \tau_{ij} + \zeta_i + \eta_j + e_{ij},$$

where $R_{ij}$ are the commuter flows between blocks $i$ and $j$, but data is reported at the district level. Hence, we write:

$$\mathbb{E}\left(\ln \pi_{ij}\right) = -\nu \overline{\tau}_{IJ} + \zeta_I + \eta_J + e_{IJ},$$

where $I, J$ are the districts of residence/employment.

- BUT: $\mathbb{E}\left(\ln \pi_{ij}\right)$ unobserved.

# Reduced form estimation

Jensen inequality gives us:

$$\ln \mathbb{E}(\pi_{ij}) \geq \mathbb{E}(\ln \pi_{ij})$$

and we can approximately estimate:

$$\ln R_{IJ} = -\nu \overline{\tau}_{IJ} + \zeta_I + \eta_J + e_{IJ},$$

where:

$$\tau_{IJ} := \frac{1}{|I||J|} \sum_{i \in I} \sum_{j \in J} \tau_{ij}.$$

# Commuting gravity equation estimation

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | ln Bilateral Commuting Probability | | | |
| Travel Time $(-\hat{\nu})$ | $-0.0482^{***}$ | $-0.0467^{***}$ | $-0.0538^{***}$ | $-0.0470^{***}$ |
| | (0.0019) | (0.0020) | (0.0020) | (0.0016) |
| Travel Time $(-\hat{\nu}_{\text{ARSW}})$ | $-0.0697^{***}$ | $-0.0702^{***}$ | $-0.0771^{***}$ | $-0.0706^{***}$ |
| Estimator | OLS | OLS | Poisson PML | Gamma PML |
| $\geq 10$ Commuters | — | Yes | Yes | Yes |
| Fixed Effects | Yes | Yes | Yes | Yes |
| $N$ | 144 | 122 | 122 | 122 |
| $R^2$ | 0.897 | 0.913 | — | — |

*Notes:* Gravity equation estimates based on Bezirke-level commuting flows and block-level travel times aggregated to the district level. Observations are bilateral pairs of 12 post-2001 Berlin Bezirke. Travel time $\bar{\tau}_{IJ}$ in minutes. Fixed effects are residence and workplace district fixed effects. Heteroscedasticity-robust SEs in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

# Discussion

- Ahlfeldt et al. (2015) TTM: modal-share weighted composite of public transport and automobile travel with headways and waiting.
- Our TTM: without automobile travel, headways and waiting.
  $\rightarrow$ Lower semi-elasticity with $\frac{\hat{\nu}_{ARSW}}{\hat{\nu}} \approx 1.4711$.
- Ahlfeldt et al. (2015):
  $\rightarrow$ "the unweighted average travel time across all possible bilateral connections with positive values of either workplace or residence employment rises from 51 minutes to 70 minutes".
  $\rightarrow \frac{51}{70} \cdot 1.4711 \approx 1.07$.

# Commuting market clearing condition

From:

$$\pi_{ij\,|\,i} = \frac{E_j w_j^{\varepsilon} d_{ij}^{-\varepsilon}}{\sum_{s=1}^{S} E_s w_s^{\varepsilon} d_{is}^{-\varepsilon}}$$

we get the commuting market clearing condition that assumes that the number of workers in block $j$ is equal to the number of workers that want to commute to $j$:

$$H_{Mj} = \sum_{i=1}^{S} \frac{\widetilde{w}_j d_{ij}^{-\varepsilon}}{\sum_{s=1}^{S} \widetilde{w}_s d_{is}^{-\varepsilon}} \cdot H_{Ri} \qquad \rightsquigarrow \qquad \widetilde{w}_j,$$

where:

- $H_{Mj}$ is the measure of workers employed in block $j$,
- $H_{Ri}$ is the measure of residents in block $i$,
- $\widetilde{w}_j := w_j^{\varepsilon} E_j$

# Identification of $\varepsilon$

We note that:

$$\sigma^2_{\ln \widetilde{w}_j} = \mathbb{E}\left(\left(\frac{1}{\varepsilon}\ln(\widetilde{w}_j) - \underbrace{\mathbb{E}\left(\frac{1}{\varepsilon}\ln(\widetilde{w}_j)\right)}_{=0}\right)^2\right) = \mathbb{E}\left(\left(\frac{1}{\varepsilon}\ln(\widetilde{w}_j)\right)^2\right)$$

and the moment condition is:

$$\mathbb{E}\left(\frac{1}{\varepsilon^2}\ln(\widetilde{w}_j)^2 - \sigma^2_{\ln w_i}\right) = 0,$$

where $\sigma^2_{\ln w_i}$ is the variance in the data. In that case, we achieve $\hat{\kappa}$ directly as:

$$\hat{\kappa} = \frac{\hat{\nu}}{\hat{\varepsilon}}.$$

# Estimates of $\kappa$

We achieve:

$$\hat{\varepsilon} = 6.92 \qquad vs \qquad \hat{\varepsilon}^{\text{ARSW}} = 6.83$$

and hence we have:

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Travel Time $(-\hat{\nu})$ | $-0.0482^{***}$ | $-0.0467^{***}$ | $-0.0538^{***}$ | $-0.0470^{***}$ |
| Travel Time $(-\hat{\nu}_{\text{ARSW}})$ | $-0.0697^{***}$ | $-0.0702^{***}$ | $-0.0771^{***}$ | $-0.0706^{***}$ |
| $\widehat{\kappa}$ | 0.00697 | 0.00672 | 0.0078 | 0.00679 |
| $\widehat{\kappa}^{\text{ARSW}}$ | 0.0102 | 0.01028 | 0.01129 | 0.01034 |
| Estimator | OLS | OLS | Poisson PML | Gamma PML |
| $\geq$ 10 Commuters | — | Yes | Yes | Yes |

# Identification Setup

**What do we observe?** Block-level data for Berlin 2006:

$$H_i^R, \quad H_j^M, \quad q_i, \quad \tau_{ij}$$

residential and workplace employment, floor prices, travel times.

**What do we want?** Two latent block fundamentals:

- $\widetilde{A}_j$ adjusted productivity of workplace $j$
- $\widetilde{B}_i$ adjusted residential amenity of block $i$

**Parameters carried forward from Task 1(b):**

$$\hat{\varepsilon}, \quad \hat{\kappa} = \hat{\nu}/\hat{\varepsilon}, \quad \alpha = 0.80, \quad \beta = 0.75$$

# Wage Fixed-Point

We start with:

$$H_{Mj} = \sum_{i=1}^{S} \frac{\widetilde{w}_j d_{ij}^{-\varepsilon}}{\sum_{s=1}^{S} \widetilde{w}_s d_{is}^{-\varepsilon}} \cdot H_{Ri} \cdot,$$

**Labor-market clearing at $j$:** $H_j^M = \sum_i H_i^R \pi_{ij}$. Substituting and solving for $\widetilde{w}_j$:

$$\widetilde{w}_j = \frac{H_j^M}{\sum_i \underbrace{\frac{H_i^R e^{-\nu\tau_{ij}}}{\sum_j \widetilde{w}_j e^{-\nu\tau_{ij}}}}_{\text{effective supply to } j}}$$

This is a fixed-point system in $\{\widetilde{w}_j\}$ because $\sum_j \widetilde{w}_j e^{-\nu\tau_{ij}}$ depends on all $\widetilde{w}_k$. Iterated contraction mapping converges to a unique solution (up to a common normalization). Wages are then $w_j = \widetilde{w}_j^{1/\varepsilon}$.

# Identifying $\widetilde{A}_j$ and $\widetilde{B}_j$

From profit-maximization and zero-profits we have:

$$\ln\left(\frac{\widetilde{A}_i}{\bar{\widetilde{A}}}\right) = (1-\alpha)\ln\left(\frac{Q_{it}}{\bar{Q}}\right) + \frac{\alpha}{\varepsilon}\ln\left(\frac{\widetilde{w}_j}{\bar{\widetilde{w}}}\right)$$

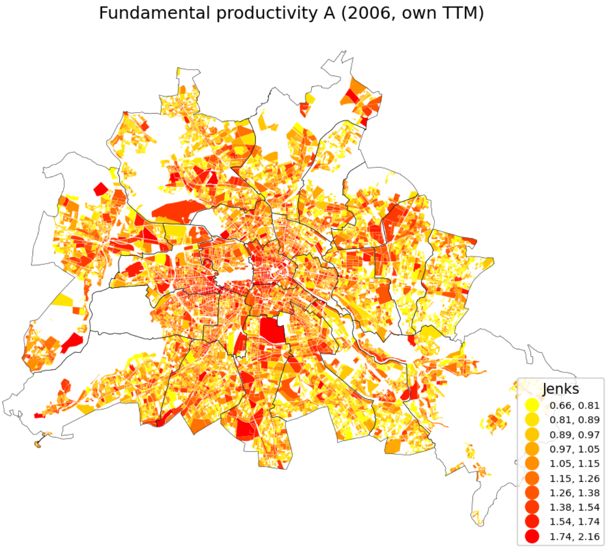
where $\widetilde{A}_i = A_i E_i^{\frac{\alpha}{\varepsilon}}$.

From residential choice prob. and population mobility we have:

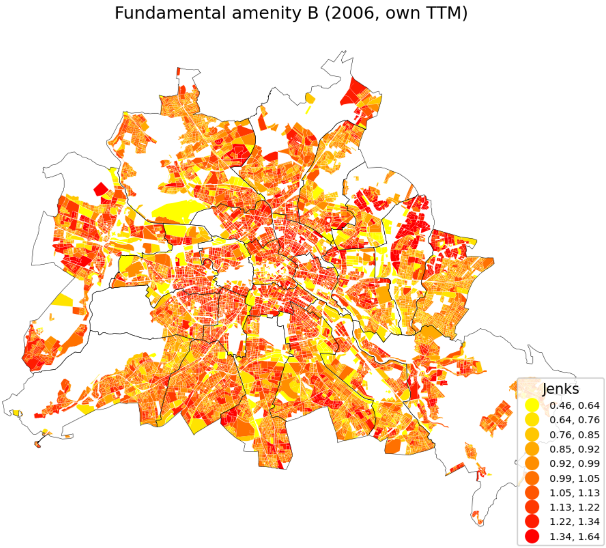
$$\ln\left(\frac{\widetilde{B}_i}{\bar{\widetilde{B}}}\right) = \frac{1}{\varepsilon}\ln\left(\frac{H_{Ri}}{\bar{H}_{Ri}}\right) + (1-\beta)\ln\left(\frac{Q_j}{\bar{Q}}\right) - \frac{1}{\varepsilon}\ln\left(\frac{W_i}{\bar{W}}\right)$$

where $\widetilde{B}_i = B_i T_i^{\frac{1}{\varepsilon}} H_{Ri}^{1-\beta}$ and $W_{it} = \sum_{s=1}^{S} \frac{\widetilde{w}_s}{e^{-\nu\tau_{is}}}$.

# Results: Fundamental Productivity



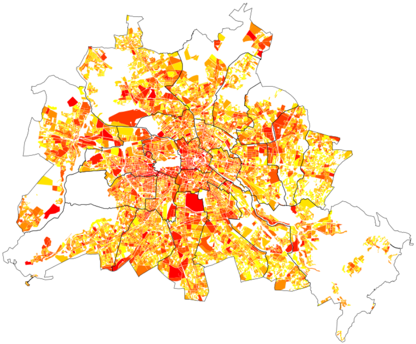
# Results: Fundamental Amenity



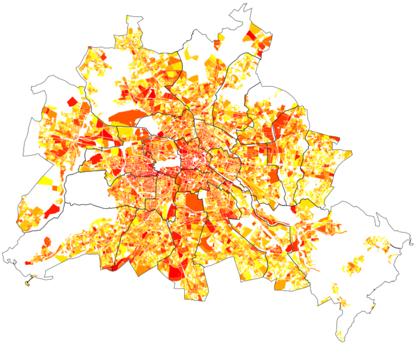
# Results: Fundamental Productivity

Fundamental productivity A — own vs ARSW travel times

Own TTM



ARSW TTM



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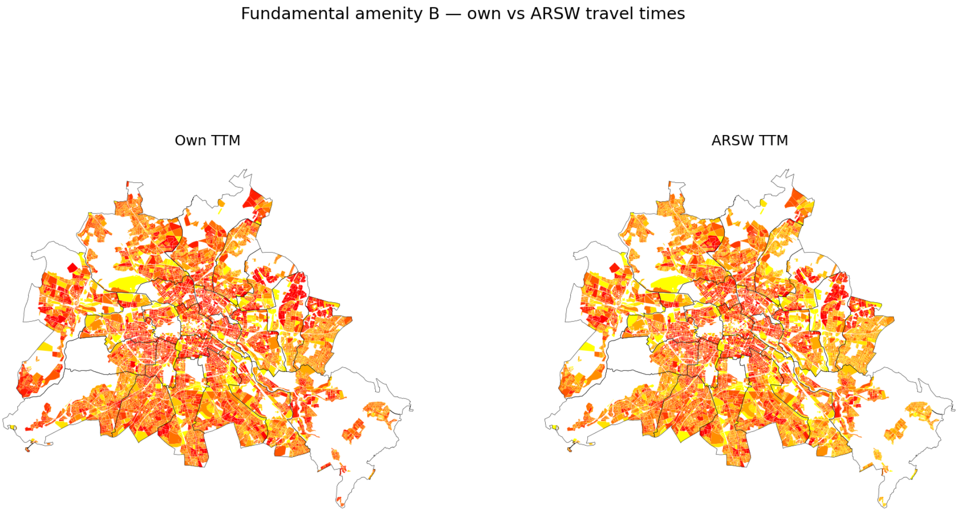

# References I

Ahlfeldt, G. M., Redding, S. J., Sturm, D. M., and Wolf, N. (2015). The economics of density: Evidence from the berlin wall. *Econometrica*, 83(6):2127–2189.